Table 1.2 Effect of dielectrics in capacitors

S. No	Dielectric is inserted	Charge *Q*	Voltage *V*	Electric field *E*	Capacitance *C*	Energy *U*
1	When the battery is disconnected	Constant	decreases	Decreases	Increases	Decreases
2	When the battery is connected	Increases	Constant	Constant	Increases	Increases

EXAMPLE 1.21

A parallel plate capacitor filled with mica having $\varepsilon_r = 5$ is connected to a 10 V battery. The area of each parallel plate is 6 cm^2 and separation distance is 6 mm. (a) Find the capacitance and stored charge.

(b) After the capacitor is fully charged, the battery is disconnected and the dielectric is removed carefully.

Calculate the new values of capacitance, stored energy and charge.

Solution

(a) The capacitance of the capacitor in the presence of dielectric is

$$C = \frac{\epsilon_r \epsilon_0 A}{d} = \frac{5 \times 8.85 \times 10^{-12} \times 6 \times 10^{-4}}{6 \times 10^{-3}}$$
$$= 44.25 \times 10^{-13}\,\text{F} = 4.425\ \text{pF}$$

The stored charge is

$$Q = CV = 44.25 \times 10^{-13} \times 10$$
$$= 442.5 \times 10^{-13}\,\text{C} = 44.25\,\text{pC}$$

The stored energy is

$$U = \frac{1}{2}CV^2 = \frac{1}{2} \times 44.25C \times 10^{-13} \times 100$$
$$= 2.21 \times 10^{-10}\,\text{J}$$

(b) After the removal of the dielectric, since the battery is already disconnected the total charge will not change. But the potential difference between the plates increases. As a result, the capacitance is decreased.

New capacitance is

$$C_0 = \frac{C}{\varepsilon_r} = \frac{4.425 \times 10^{-12}}{5}$$
$$= 0.885 \times 10^{-12}\,\text{F} = 0.885\,\text{pF}$$

The stored charge remains same and 44.25 pC. Hence newly stored energy is

$$U_0 = \frac{Q^2}{2C_0} = \frac{Q^2 \epsilon_r}{2C} = \epsilon_r U$$
$$= 5 \times 2.21 \times 10^{-10}\,\text{J} = 11.05 \times 10^{-10}\,\text{J}$$

The increased energy is

$$\Delta U = (11.05 - 2.21) \times 10^{-10}\,\text{J} = 8.84 \times 10^{-10}\,\text{J}$$

When the dielectric is removed, it experiences an inward pulling force due to the plates. To remove the dielectric, an external agency has to do work on the dielectric which is stored as additional energy. This is the source for the extra energy 8.84×10^{-10} J.

1.8.5 Capacitor in series and parallel

(i) Capacitor in series

Consider three capacitors of capacitance C_1, C_2 and C_3 connected in series with a battery of voltage V as shown in the Figure 1.58 (a).

As soon as the battery is connected to the capacitors in series, the electrons of charge

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